

CHAPTER 18

SEISMIC DESIGN REQUIREMENTS FOR STRUCTURES WITH DAMPING SYSTEMS

18.1 GENERAL

18.1.1 Scope. Every structure with a damping system and every portion thereof shall be designed and constructed in accordance with the requirements of this standard as modified by this chapter. Where damping devices are used across the isolation interface of a seismically isolated structure, displacements, velocities, and accelerations shall be determined in accordance with Chapter 17.

18.1.2 Definitions. The following definitions apply only to the structures with damping system provisions of Chapter 18 and are in addition to the definitions presented in Chapter 11:

DAMPING DEVICE: A flexible structural element of the damping system that dissipates energy caused by relative motion of each end of the device. Damping devices include all pins, bolts, gusset plates, brace extensions, and other components required to connect damping devices to the other elements of the structure. Damping devices are classified as either displacement-dependent or velocity-dependent, or a combination thereof, and are permitted to be configured to act in either a linear or nonlinear manner.

DAMPING SYSTEM: The collection of structural elements that includes all the individual damping devices, all structural elements or bracing required to transfer forces from damping devices to the base of the structure, and the structural elements required to transfer forces from damping devices to the seismic force-resisting system.

DISPLACEMENT-DEPENDENT DAMPING DEVICE: The force response of a displacement-dependent damping device is primarily a function of the relative displacement between each end of the device. The response is substantially independent of the relative velocity between each of the device and/or the excitation frequency.

FORCE-CONTROLLED ELEMENTS: Element actions for which reliable inelastic deformation capacity is not achievable without critical strength decay.

VELOCITY-DEPENDENT DAMPING DEVICE: The force-displacement relation for a velocity-dependent damping device is primarily a function of the relative velocity between each end of the device and could also be a function of the relative displacement between each end of the device.

18.1.3 Symbols. Symbols presented in this section apply only to the structures with damping system provisions of Chapter 18 and are in addition to the symbols presented in Chapter 11.

B_{1D} = numerical coefficient as set forth in Table 18.7-1 for effective damping equal to β_{mD} ($m = 1$) and period of structure equal to T_{1D}

B_{1E} = numerical coefficient as set forth in Table 18.7-1 for the effective damping equal to $\beta_I + \beta_{V1}$ and period equal to T_1

B_{1M} = numerical coefficient as set forth in Table 18.7-1 for effective damping equal to β_{mM} ($m = 1$) and period of structure equal to T_{1M}

B_{mD} = numerical coefficient as set forth in Table 18.7-1 for effective damping equal to β_{mI} and period of structure equal to T_m

B_{mM} = numerical coefficient as set forth in Table 18.7-1 for effective damping equal to β_{mM} and period of structure equal to T_m

B_R = numerical coefficient as set forth in Table 18.7-1 for effective damping equal to β_R and period of structure equal to T_R

B_{V+I} = numerical coefficient as set forth in Table 18.7-1 for effective damping equal to the sum of viscous damping in the fundamental mode of vibration of the structure in the direction of interest, β_{Vm} ($m = 1$), plus inherent damping, β_I , and period of structure equal to T_1

C_{mFD} = force coefficient as set forth in Table 18.7-2

C_{mFV} = force coefficient as set forth in Table 18.7-3

C_{S1} = seismic response coefficient of the fundamental mode of vibration of the structure in the direction of interest, Section 18.7.1.2.4 or 18.7.2.2.4 ($m = 1$)

C_{Sm} = seismic response coefficient of the m th mode of vibration of the structure in the direction of interest, Section 18.7.1.2.4 ($m = 1$) or Section 18.7.1.2.6 ($m > 1$)

C_{SR} = seismic response coefficient of the residual mode of vibration of the structure in the direction of interest, Section 18.7.2.2.8

D_{1D} = fundamental mode design displacement at the center of rigidity of the roof level of the structure in the direction under consideration, Section 18.7.2.3.2

D_{1M} = fundamental mode MCE_R displacement at the center of rigidity of the roof level of the structure in the direction under consideration, Section 18.7.2.3.5

D_{mD} = design displacement at the center of rigidity of the roof level of the structure caused by the m th mode of vibration in the direction under consideration, Section 18.7.1.3.2

D_{mM} = MCE_R displacement at the center of rigidity of the roof level of the structure caused by the m th mode of vibration in the direction under consideration, Section 18.7.1.3.5

D_{RD} = residual mode design displacement at the center of rigidity of the roof level of the structure in the direction under consideration, Section 18.7.2.3.2

D_{RM} = residual mode MCE_R displacement at the center of rigidity of the roof level of the structure in the direction under consideration, Section 18.7.2.3.5

D_Y = displacement at the center of rigidity of the roof level of the structure at the effective yield point of the seismic force-resisting system, Section 18.7.3.3
 E_{loop} = area of one load-displacement hysteresis loop, Section 18.6.1.5.1
 f_i = lateral force at level i of the structure distributed approximately in accordance with Section 12.8.3, Section 18.7.2.2.3
 F_{i1} = inertial force at level i (or mass point i) in the fundamental mode of vibration of the structure in the direction of interest, Section 18.7.2.2.9
 F_{im} = inertial force at level i (or mass point i) in the m th mode of vibration of the structure in the direction of interest, Section 18.7.1.2.7
 F_{iR} = inertial force at level i (or mass point i) in the residual mode of vibration of the structure in the direction of interest, Section 18.7.2.2.9
 h_i = height above the base to level i , Section 18.7.2.2.3
 h_n = structural height, Section 18.7.2.2.3
 q_H = hysteresis loop adjustment factor as determined in Section 18.7.3.2.2.1
 Q_{DSD} = force in an element of the damping system required to resist design seismic forces of displacement-dependent damping devices, Section 18.7.4.5
 Q_E = seismic design force in each element of the damping system, Section 18.7.4.5
 Q_{mDSV} = force in an element of the damping system required to resist design seismic forces of velocity-dependent damping devices caused by the m th mode of vibration of the structure in the direction of interest, Section 18.7.4.5
 Q_{mSFRS} = force in an element of the damping system equal to the design seismic force of the m th mode of vibration of the structure in the direction of interest, Section 18.7.4.5
 T_1 = fundamental period of the structure in the direction under consideration
 T_{1D} = effective period, in seconds, of the fundamental mode of vibration of the structure at the design displacement in the direction under consideration, as prescribed by Section 18.7.1.2.5 or 18.7.2.2.5
 T_{1M} = effective period, in seconds, of the fundamental mode of vibration of the structure at the MCE_R displacement in the direction under consideration, as prescribed by Section 18.7.1.2.5 or 18.7.2.2.5
 T_m = period, in seconds, of the m th mode of vibration of the structure in the direction under consideration, Section 18.7.1.2.6
 T_R = period, in seconds, of the residual mode of vibration of the structure in the direction under consideration, Section 18.7.2.2.7
 V = seismic base shear in the direction of interest, Section 18.2.1.1
 V_1 = design value of the seismic base shear of the fundamental mode in a given direction of response, as determined in Section 18.7.2.2.1 and Section 18.7.2.2.2,
 V_m = design value of the seismic base shear of the m th mode of vibration of the structure in the direction of interest, Section 18.7.1.2.2
 V_{min} = minimum allowable value of base shear permitted for design of the seismic force-resisting system of the structure in the direction of interest, Section 18.2.1.1
 V_R = design value of the seismic base shear of the residual mode of vibration of the structure in a given direction, as determined in Section 18.7.2.2.6

w_i = effective seismic weight of the i th floor of the structure, Section 18.7.1.2.2
 $\bar{W}_1$ = effective fundamental mode seismic weight determined in accordance with Eq. (18.7-2b) for $m = 1$
 $\bar{W}_m$ = effective seismic weight of the m th mode of vibration of the structure, Section 18.7.1.2.2
 W_m = maximum strain energy in the m th mode of vibration of the structure in the direction of interest at modal displacements, δ_{im} , Section 18.7.3.2.2.1
 W_{mj} = work done by j th damping device in one complete cycle of dynamic response corresponding to the m th mode of vibration of the structure in the direction of interest at modal displacements, δ_{im} , Section 18.7.3.2.2.1
 $\bar{W}_R$ = effective residual mode seismic weight determined in accordance with Eq. (18.7-30)
 α = velocity exponent relating damping device force to damping device velocity
 β_{HD} = component of effective damping of the structure in the direction of interest caused by postyield hysteretic behavior of the seismic force-resisting system and elements of the damping system at effective ductility demand μ_D , Section 18.7.3.2.2
 β_{HM} = component of effective damping of the structure in the direction of interest caused by postyield hysteretic behavior of the seismic force-resisting system and elements of the damping system at effective ductility demand, μ_M , Section 18.7.3.2.2
 β_I = component of effective damping of the structure caused by the inherent dissipation of energy by elements of the structure, at or just below the effective yield displacement of the seismic force-resisting system, Section 18.7.3.2.1
 β_{mD} = total effective damping of the m th mode of vibration of the structure in the direction of interest at the design displacement, Section 18.7.3.2
 β_{mM} = total effective damping of the m th mode of vibration of the structure in the direction of interest at the MCE_R displacement, Section 18.7.3.2
 β_R = total effective damping in the residual mode of vibration of the structure in the direction of interest, calculated in accordance with Section 18.7.3.2 (using $\mu_D = 1.0$ and $\mu_M = 1.0$)
 β_{Vm} = component of effective damping of the m th mode of vibration of the structure in the direction of interest caused by viscous dissipation of energy by the damping system, at or just below the effective yield displacement of the seismic force-resisting system, Section 18.7.3.2.3
 Γ_1 = participation factor of the fundamental mode of vibration of the structure in the direction of interest, Section 18.7.1.2.3 or 18.7.2.2.3 ($m = 1$)
 Γ_m = participation factor in the m th mode of vibration of the structure in the direction of interest, Section 18.7.1.2.3
 Γ_R = participation factor of the residual mode of vibration of the structure in the direction of interest, Section 18.7.2.2.7
 δ_i = elastic deflection of level i of the structure caused by applied lateral force, f_i , Section 18.7.2.2.3
 δ_{i1D} = fundamental mode design deflection of level i at the center of rigidity of the structure in the direction under consideration, Section 18.7.2.3.1
 δ_{iD} = total design deflection of level i at the center of rigidity of the structure in the direction under consideration, Section 18.7.2.3

δ_{iM} = total MCE_R deflection of level i at the center of rigidity of the structure in the direction under consideration, Section 18.7.2.3
 δ_{im} = deflection of level i in the m th mode of vibration at the center of rigidity of the structure in the direction under consideration, Section 18.7.3.2.3
 δ_{imD} = design deflection of level i in the m th mode of vibration at the center of rigidity of the structure in the direction under consideration, Section 18.7.1.3.1
 δ_{iRD} = residual mode design deflection of level i at the center of rigidity of the structure in the direction under consideration, Section 18.7.2.3.1
 Δ_{1D} = design story drift caused by the fundamental mode of vibration of the structure in the direction of interest, Section 18.7.2.3.3
 Δ_D = total design story drift of the structure in the direction of interest, Section 18.7.2.3.3
 Δ_M = total MCE_R story drift of the structure in the direction of interest, Section 18.7.2.3
 Δ_{mD} = design story drift caused by the m th mode of vibration of the structure in the direction of interest, Section 18.7.1.3.3
 Δ_{RD} = design story drift caused by the residual mode of vibration of the structure in the direction of interest, Section 18.7.2.3.3
 $\lambda_{(ae, \max)}$ = factor to represent possible variation in damper properties above the tested values caused by aging and environmental effects; this is a multiple of all the individual aging and environmental effects, Section 18.2.4.5
 $\lambda_{(ae, \min)}$ = factor to represent possible variation in damper properties below the tested values caused by aging and environmental effects; this is a multiple of all the individual aging and environmental effects, Section 18.2.4.5
 $\lambda_{\max}$ = factor to represent possible total variation in damper properties above the nominal properties, Section 18.2.4.5
 $\lambda_{\min}$ = factor to represent possible total variation in damper properties below the nominal properties, Section 18.2.4.5
 $\lambda_{(\text{spec}, \max)}$ = factor to represent permissible variation in production damper nominal properties above those assumed in design, Section 18.2.4.5
 $\lambda_{(\text{spec}, \min)}$ = factor to represent permissible variation in production damper nominal properties below those assumed in design, Section 18.2.4.5
 $\lambda_{(\text{test}, \max)}$ = factor to represent possible variations in damper properties above the nominal values obtained from the prototype tests; this is a multiple of all the testing effects, Section 18.2.4.5
 $\lambda_{(\text{test}, \min)}$ = factor to represent possible variations in damper properties below the nominal values obtained from the prototype tests; this is a multiple of all the testing effects, Section 18.2.4.5
 μ = effective ductility demand on the seismic force-resisting system in the direction of interest
 μ_D = effective ductility demand on the seismic force-resisting system in the direction of interest caused by the design earthquake ground motions, Section 18.7.3.3
 μ_M = effective ductility demand on the seismic force-resisting system in the direction of interest caused by the MCE_R ground motions, Section 18.7.3.3

$\mu_{\max}$ = maximum allowable effective ductility demand on the seismic force-resisting system caused by the design earthquake ground motions, Section 18.7.3.4
 φ_{i1} = displacement amplitude at level i of the fundamental mode of vibration of the structure in the direction of interest, normalized to unity at the roof level, Section 18.7.2.2.3
 φ_{im} = displacement amplitude at level i of the m th mode of vibration of the structure in the direction of interest, normalized to unity at the roof level, Section 18.7.1.2.2
 φ_{iR} = displacement amplitude at level i of the residual mode of vibration of the structure in the direction of interest normalized to unity at the roof level, Section 18.7.2.2.7
 ∇_{1D} = design story velocity caused by the fundamental mode of vibration of the structure in the direction of interest, Section 18.7.2.3.4
 ∇_D = total design story velocity of the structure in the direction of interest, Section 18.7.1.3.4
 ∇_M = total MCE_R story velocity of the structure in the direction of interest, Section 18.7.2.3
 ∇_{mD} = design story velocity caused by the m th mode of vibration of the structure in the direction of interest, Section 18.7.1.3.4
 ∇_{RD} = design story velocity caused by the residual mode of vibration of the structure in the direction of interest, Section 18.7.2.3.4

18.2 GENERAL DESIGN REQUIREMENTS

18.2.1 System Requirements. Design of the structure shall consider the basic requirements for the seismic force-resisting system and the damping system as defined in the following sections. The seismic force-resisting system shall have the required strength to meet the forces defined in Section 18.2.1.1. The combination of the seismic force-resisting system and the damping system is permitted to be used to meet the drift requirement.

18.2.1.1 Seismic Force-Resisting System. Structures that contain a damping system shall have a seismic force-resisting system that, in each lateral direction, conforms to one of the types indicated in Table 12.2-1.

The design of the seismic force-resisting system in each direction shall satisfy the minimum base shear requirements of this section and the requirements of Section 18.4 if the nonlinear response history procedure of Section 18.3 is used, or Section 18.7.4 if either the response spectrum procedure of Section 18.7.1 or the equivalent lateral force procedure of Section 18.7.2 is used.

The seismic base shear used for design of the seismic force-resisting system shall not be less than $V_{\min}$, where $V_{\min}$ is determined as the greater of the values computed using Eqs. (18.2-1) and (18.2-2):

$$V_{\min} = \frac{V}{B_{V+1}} \quad (18.2-1)$$

$$V_{\min} = 0.75V \quad (18.2-2)$$

where

V = seismic base shear in the direction of interest, determined in accordance with Section 12.8, and

B_{V+I} = numerical coefficient as set forth in Table 18.7-1 for effective damping equal to the sum of viscous damping in the fundamental mode of vibration of the structure in the direction of interest, $\beta_{Vm}(m=1)$, plus inherent damping, β_I , and period of structure equal to T_1 .

EXCEPTION: The seismic base shear used for design of the seismic force-resisting system shall not be taken as less than 1.0V if either of the following conditions apply:

1. In the direction of interest, the damping system has fewer than two damping devices on each floor level, configured to resist torsion.
2. The seismic force-resisting system has horizontal irregularity Type 1b (Table 12.3-1) or vertical irregularity Type 1b (Table 12.3-2).

18.2.1.2 Damping System. Damping devices and all other components required to connect damping devices to the other elements of the structure shall be designed to remain elastic for MCE_R loads. Other elements of the damping system are permitted to have inelastic response at MCE_R if it is shown by analysis or test that inelastic response of these elements would not adversely affect damping system function. If either the response spectrum procedure of Section 18.7.1 or the equivalent lateral force procedure of Section 18.7.2 is used, the inelastic response shall be limited in accordance with the requirements of Section 18.7.4.6.

Force-controlled elements of the damping system shall be designed for seismic forces that are increased by 20% from those corresponding to average MCE_R response.

18.2.2 Seismic Ground Motion Criteria

18.2.2.1 Design Earthquake and MCE_R Response Spectra. The response spectrum requirements of Sections 11.4.5 and 11.4.6 are permitted to be used to determine the design earthquake and MCE_R response spectra for the site of interest. The site-specific ground motion procedures set forth in Chapter 21 are also permitted to be used to determine ground motions for any structure with a damping system.

For any structure with a damping system located on a Site Class F site, site response analysis shall be performed in accordance with Section 21.1.

18.2.2.2 Design Earthquake and MCE_R Ground Motion Records. Where nonlinear response history analysis procedures are used, design earthquake and MCE_R ground motion suites shall each consist of not less than seven pairs of horizontal acceleration components selected and scaled from individual recorded events that have magnitudes, fault distance, and source mechanisms that are consistent with those that control the design earthquake and MCE_R events. Amplitude or spectral matching is permitted to scale the ground motions. Where the required number of recorded ground motion pairs is not available, simulated ground motion pairs are permitted to make up the total number required.

For each pair of horizontal ground motion components, a square root of the sum of the squares (SRSS) spectrum shall be constructed by taking the SRSS of the 5% damped response spectra for the scaled components (when amplitude scaling is used, an identical scale factor is applied to both components of a pair). For both the design earthquake and the MCE_R suites, each pair of motions shall be scaled such that in the period range from $0.2T_{1D}$ to $1.25T_{1M}$, the average of the SRSS spectra from all horizontal component pairs does not fall below the corresponding ordinate of the response spectra used in the design, determined in accordance with Section 18.2.2.1.

For records that are spectrally matched, for both the design earthquake and the MCE_R , each pair of motions shall be scaled such that in the period range from $0.2T_{1D}$ to $1.25T_{1M}$, the response spectrum of one component of the pair is at least 90% of the corresponding ordinate of the response spectrum used in the design, determined in accordance with Section 18.2.2.1.

For sites within 3 mi (5 km) of the active fault that controls the hazard, spectral matching shall not be used unless the pulse characteristics of the near-field ground motions are included in the site-specific response spectra, and pulse characteristics, when present in individual ground motions, are retained after the matching process has been completed.

At sites within 3 mi (5 km) of the active fault that controls the hazard, for both the design earthquake and the MCE_R suites, each pair of components shall be rotated to the fault-normal and fault-parallel directions of the causative fault and shall be scaled so that the average spectrum of the fault-normal components is not less than the design or MCE_R response spectrum, as appropriate, and the average spectrum of the fault-parallel components is not less than 50% of the design or MCE_R response spectrum, as appropriate, for the period range from $0.2T_{1D}$ to $1.25T_{1M}$.

18.2.3 Procedure Selection. Structures with a damping system provided for seismic resistance shall be analyzed and designed using the nonlinear response history procedure of Section 18.3.

EXCEPTION: It shall be permitted to analyze and design the structure using the response spectrum procedure of Section 18.7.1 subject to the limits of Section 18.2.3.1 or the equivalent lateral force procedure of Section 18.7.2 subject to the limits of Section 18.2.3.2.

18.2.3.1 Response Spectrum Procedure. The response spectrum procedure of Section 18.7.1 is permitted to be used for analysis and design provided that all of the following conditions apply:

1. In each principal direction, the damping system has at least two damping devices in each story, configured to resist torsion.
2. The total effective damping of the fundamental mode, $\beta_{mD}(m=1)$, of the structure in the direction of interest is not greater than 35% of critical.
3. The S_1 value for the site is less than 0.6.

18.2.3.2 Equivalent Lateral Force Procedure. The equivalent lateral force procedure of Section 18.7.2 is permitted to be used for analysis and design provided that all of the following conditions apply:

1. In each principal direction, the damping system has at least two damping devices in each story, configured to resist torsion.
2. The total effective damping of the fundamental mode, $\beta_{mD}(m=1)$, of the structure in the direction of interest is not greater than 35% of critical.
3. The seismic force-resisting system does not have horizontal irregularity Type 1a or 1b (Table 12.3-1) or vertical irregularity Type 1a, 1b, 2, or 3 (Table 12.3-2).
4. Floor diaphragms are rigid as defined in Section 12.3.1.
5. The height of the structure above the base does not exceed 100 ft (30 m).
6. The S_1 value for the site is less than 0.6.

18.2.4 Damping System

18.2.4.1 Device Design. The design, construction, and installation of damping devices shall be based on response to MCE_R ground motions and consideration of all of the following:

1. Low-cycle, large-displacement degradation caused by seismic loads.
2. High-cycle, small-displacement degradation caused by wind, thermal, or other cyclic loads.
3. Forces or displacements caused by gravity loads.
4. Adhesion of device parts caused by corrosion or abrasion, biodegradation, moisture, or chemical exposure.
5. Exposure to environmental conditions, including, but not limited to, temperature, humidity, moisture, radiation (e.g., ultraviolet light), and reactive or corrosive substances (e.g., saltwater).

Devices using bimetallic interfaces subject to cold welding of the sliding interface shall be prohibited from use in a damping system.

Damping devices subject to failure by low-cycle fatigue shall resist wind forces without slip, movement, or inelastic cycling.

The design of damping devices shall incorporate the range of thermal conditions, device wear, manufacturing tolerances, and other effects that cause device properties to vary during the design life of the device in accordance with Section 18.2.4.4. Ambient temperature shall be the normal in-service temperature of the damping device. The design temperature range shall cover the annual minimum and maximum in-service temperatures of the damping device.

18.2.4.2 Multiaxis Movement. Connection points of damping devices shall provide sufficient articulation to accommodate simultaneous longitudinal, lateral, and vertical displacements of the damping system.

18.2.4.3 Inspection and Periodic Testing. Means of access for inspection and removal of all damping devices shall be provided.

The registered design professional (RDP) responsible for design of the structure shall establish an inspection, maintenance, and testing schedule for each type of damping device to ensure that the devices respond in a dependable manner throughout their design life. The degree of inspection and testing shall reflect the established in-service history of the damping devices and the likelihood of change in properties over the design life of the devices.

18.2.4.4 Nominal Design Properties. Nominal design properties for energy-dissipation devices shall be established from either project-specific prototype test data or prior prototype tests on devices of similar type and size. The nominal design properties shall be based on data from prototype tests specified in Section 18.6.1.2 (2) and determined by Section 18.6.1.4 (2). These nominal design properties shall be modified by property variation or lambda (λ) factors as specified in Section 18.2.4.5.

18.2.4.5 Maximum and Minimum Damper Properties. Maximum and minimum property modification (λ) factors shall be established in accordance with Eqs. (18.2-3a) and (18.2-3b) for each device by the RDP and used in analysis and design to account for the variation from nominal properties.

$$\lambda_{\max} = [(1 + (0.75 \times (\lambda_{(ae, \max)} - 1))) \times \lambda_{(test, \max)} \times \lambda_{(spec, \max)}] \geq 1.2 \quad (18.2-3a)$$

$$\lambda_{\min} = [(1 - (0.75 \times (1 - \lambda_{(ae, \min)}))) \times \lambda_{(test, \min)} \times \lambda_{(spec, \min)}] \leq 0.85 \quad (18.2-3b)$$

where

$\lambda_{(ae, \max)}$ = factor to represent possible variation in damper properties above the nominal values caused by aging and environmental effects; this is a multiple of all the individual aging and environmental effects;

$\lambda_{(ae, \min)}$ = factor to represent possible variation in damper properties below the nominal values caused by aging and environmental effects; this is a multiple of all the individual aging and environmental effects;

$\lambda_{(test, \max)}$ = factor to represent possible variations in damper properties above the nominal values obtained from the prototype tests; this is a multiple of all the testing effects;

$\lambda_{(test, \min)}$ = factor to represent possible variations in damper properties below the nominal values obtained from the prototype tests; this is a multiple of all the testing effects;

$\lambda_{(spec, \max)}$ = factor established by the RDP to represent permissible variation in production damper properties above the nominal values; and

$\lambda_{(spec, \min)}$ = factor established by the RDP to represent permissible variation in production damper properties below the nominal values.

EXCEPTION: With test data reviewed by the RDP and accepted by peer review, it is permitted to use $\lambda_{\max}$ less than 1.2 and $\lambda_{\min}$ greater than 0.85.

Maximum and minimum analysis and design properties for each device shall be determined in accordance with Eqs. (18.2-4a) and (18.2-4b) for each modeling parameter as follows:

$$\text{Maximum Design Property} = \text{Nominal Design Property} \times \lambda_{\max} \quad (18.2-4a)$$

$$\text{Minimum Design Property} = \text{Nominal Design Property} \times \lambda_{\min} \quad (18.2-4b)$$

A maximum and minimum analysis and design property shall be established for each modeling parameter as necessary for the selected method of analysis. Maximum velocity coefficients, stiffness, strength, and energy dissipation shall be considered together as the maximum analysis and design case, and minimum velocity coefficients, strength, stiffness, and energy dissipation shall be considered together as the minimum analysis and design case.

Separate maximum and minimum properties shall be established for loads and displacements corresponding to the design level conditions and the MCE_R conditions.

18.2.4.6 Damping System Redundancy. If fewer than four energy-dissipation devices are provided in any story of a building in either principal direction, or fewer than two devices are located on each side of the center of stiffness of any story in either principal direction, all energy-dissipation devices shall be capable of sustaining displacements equal to 130% of the maximum calculated displacement in the device under MCE_R . A velocity-dependent device shall be capable of sustaining the force and displacement associated with a velocity equal to 130% of the maximum calculated velocity for that device under MCE_R .

18.3 NONLINEAR RESPONSE HISTORY PROCEDURE

The stiffness and damping properties of the damping devices used in the models shall be based on or verified by testing of the damping devices as specified in Section 18.6. The nonlinear force-velocity-displacement characteristics of damping devices shall be modeled, as required, to explicitly account for device dependence on frequency, amplitude, and duration of seismic loading.

A nonlinear response history analysis shall use a mathematical model of the seismic force-resisting system and the damping system as provided in this section. The model shall directly account for the nonlinear hysteretic behavior of all members and connections undergoing inelastic behavior, in a manner consistent with applicable laboratory test data. Test data shall not be extrapolated beyond tested deformation levels. If the analysis results indicate that degradation in element strength or stiffness can occur, the hysteretic models shall include these effects.

EXCEPTION: If the calculated force in an element of the seismic force-resisting system or the damping system does not exceed 1.5 times its expected strength using strength reduction factor $\phi = 1$, that element is permitted to be modeled as linear.

For sites identified as near-fault, each pair of horizontal ground motion components shall be rotated to the fault-normal and fault-parallel directions of the causative faults and applied to the mathematical model in such orientation.

For all other sites, individual pairs of horizontal ground motion components need not be applied to the mathematical model in multiple orientations.

Inherent damping of the structure shall not be taken as greater than 3% of critical unless test data consistent with levels of deformation at or just below the effective yield displacement of the seismic force-resisting system support higher values.

Analysis shall be performed at both the design earthquake and at the MCE_R earthquake levels. The design earthquake analysis need not include the effects of accidental eccentricity. Results from the design earthquake analysis shall be used to design the seismic force-resisting system. Results from the MCE_R analysis shall be used to design the damping system.

18.3.1 Damping Device Modeling. Mathematical models of displacement-dependent damping devices shall include the hysteretic behavior of the devices consistent with test data and accounting for all significant changes in strength, stiffness, and hysteretic loop shape. Mathematical models of velocity-dependent damping devices shall include the velocity coefficient consistent with test data. If damping device properties change with time and/or temperature, such behavior shall be modeled explicitly. The flexible elements of damping devices connecting damper units to the structure shall be included in the model.

EXCEPTION: If the properties of the damping devices are expected to change during the duration of the response history analysis, the dynamic response is permitted to be enveloped by the maximum and minimum device properties from Section 18.2.4.5. All these limit cases for variable device properties shall satisfy the same conditions as if the time-dependent behavior of the devices were explicitly modeled.

18.3.2 Accidental Mass Eccentricity. Inherent eccentricity resulting from lack of symmetry in mass and stiffness shall be accounted for in the MCE_R analysis. In addition, accidental eccentricity consisting of displacement of the center of mass

from the computed location by an amount equal to 5% of the diaphragm dimension separately in each of two orthogonal directions at each diaphragm level shall be accounted for in the analysis.

EXCEPTION: It is permitted to account for the effects of accidental eccentricity through the establishment of amplification factors on forces, drifts, and deformations that permit results determined from an analysis using only the computed center-of-mass configuration to be scaled to bound the results of all the mass-eccentric cases.

18.3.3 Response Parameters. Maximum values of each response parameter of interest shall be calculated for each ground motion used for the response history analysis. Response parameters shall include the forces, displacements, and velocities (in the case of velocity-dependent devices) in each discrete damping device. The average value of a response parameter of interest across the suite of design earthquake or MCE_R motions is permitted to be used for design.

18.4 SEISMIC LOAD CONDITIONS AND ACCEPTANCE CRITERIA FOR NONLINEAR RESPONSE HISTORY PROCEDURE

For the nonlinear response history procedure of Section 18.3, the seismic force-resisting system, damping system, loading conditions, and acceptance criteria for response parameters of interest shall conform with the requirements of the following subsections.

18.4.1 Seismic Force-Resisting System. The seismic force-resisting system shall satisfy the strength requirements of Section 12.2.1 using both

1. the seismic base shear, V_{min} , as given by Section 18.2.1.1, and
2. the demands from the design earthquake nonlinear response history analysis.

The story drifts shall be determined using the MCE_R ground motions with the combined model of the seismic force-resisting system and the damping system. Accidental eccentricity shall be included.

The maximum drift at MCE_R shall not exceed 3%, nor the drift limits specified in Table 12.12-1 times the smaller of $1.5R/C_d$ and 1.9. C_d and R shall be taken from Table 12.2-1 for the building framing under consideration.

18.4.2 Damping System. The damping devices and their connections shall be sized to resist the forces, displacements, and velocities from the MCE_R ground motions. Force-controlled elements of the damping system shall be designed for seismic forces that are increased by 20% from those corresponding to average MCE_R response.

18.4.3 Combination of Load Effects. The effects on the damping system caused by gravity loads and seismic forces shall be combined in accordance with Section 12.4 using the effect of horizontal seismic forces, Q_E , except that Q_E shall be determined in accordance with the MCE_R analysis. When load combinations are used that include live loading, it is permitted to use a load factor of 25% on live load for nonlinear response history analysis. The redundancy factor, ρ , shall be taken equal to 1.0 in all cases, and the seismic load effect including overstrength of Section 12.4.3 need not apply to the design of the damping system.

18.4.4 Acceptance Criteria for the Response Parameters of Interest. The damping system components shall be evaluated by the strength design criteria of this standard using the seismic forces and seismic loading conditions determined from the MCE_R nonlinear response history analyses and strength reduction factor $\phi = 1.0$.

18.5 DESIGN REVIEW

An independent design review of the damping system and related test programs shall be performed by one or more individuals possessing knowledge of the following items; a minimum of one reviewer shall be a registered design professional. Damping system design review shall include, but need not be limited to, all of the following:

1. Project design criteria including site-specific spectra and ground motion histories;
2. Preliminary design of the seismic force-resisting system and the damping system, including selection of the devices and their design parameters;
3. Review of manufacturer test data and property modification factors for the manufacturer and device selected;
4. Prototype testing program (Section 18.6.1);
5. Final design of the entire structural system and supporting analyses, including modeling of the damping devices for response history analysis if performed; and
6. Damping device production testing program (Section 18.6.2).

18.6 TESTING

The force-velocity-displacement relationships and damping properties assumed as the damping device nominal design properties in Section 18.2.4.4 shall be confirmed by the tests conducted in accordance with Section 18.6.1 or shall be based on prior tests of devices meeting the similarity requirements of Section 18.6.1.3.

The prototype tests specified in Section 18.6.1 shall be conducted to confirm the force-velocity-displacement properties of the damping devices assumed for analysis and design and to demonstrate the robustness of individual devices under seismic excitation. These tests shall be conducted prior to production of devices for construction.

The production testing requirements are specified in Section 18.6.2.

Device nominal properties determined from the prototype testing shall meet the acceptance criteria established using $\lambda_{(spec, max)}$ and $\lambda_{(spec, min)}$ from Section 18.2.4.5. These criteria shall account for likely variations in material properties.

Device nominal properties determined from the production testing of Section 18.6.2 shall meet the acceptance criteria established using $\lambda_{(spec, max)}$ and $\lambda_{(spec, min)}$ from Section 18.2.4.5.

The fabrication and quality control procedures used for all prototype and production devices shall be identical. These procedures shall be approved by the RDP prior to the fabrication of prototype devices.

18.6.1 Prototype Tests. The following tests shall be performed separately on two full-size damping devices of each type and size used in the design, in the order listed as follows.

Representative sizes of each type of device are permitted to be used for prototype testing, provided that both of the following conditions are met:

1. Fabrication and quality control procedures are identical for each type and size of device used in the structure.

2. Prototype testing of representative sizes is approved by the RDP responsible for design of the structure.

Test specimens shall not be used for construction, unless they are approved by the RDP responsible for design of the structure and meet the requirements for prototype and production tests.

18.6.1.1 Data Recording. The force-deflection relationship for each cycle of each test shall be recorded electronically.

18.6.1.2 Sequence and Cycles of Testing. For all of the following test sequences, each damping device shall be subjected to gravity load effects and thermal environments representative of the installed condition. For seismic testing, the displacement in the devices calculated for the MCE_R ground motions, termed herein as the maximum device displacement, shall be used.

Prior to the sequence of prototype tests defined in this section, a production test in accordance with Section 18.6.2 shall be performed, and data from this test shall be used as a baseline for comparison with subsequent tests on production dampers.

1. Each damping device shall be subjected to the number of cycles expected in the design windstorm, but not less than 2,000 continuous fully reversed cycles of wind load. Wind load shall be at amplitudes expected in the design windstorm and shall be applied at a frequency equal to the inverse of the fundamental period of the structure, $1/T_1$.

It is permitted to use alternate loading protocols, representative of the design windstorm, that apportion the total wind displacement into its expected static, pseudostatic, and dynamic components.

EXCEPTION: Damping devices need not be subjected to these tests if they are not subject to wind-induced forces or displacements or if the design wind force is less than the device yield or slip force.

2. Each damping device shall be brought to ambient temperature and loaded with the following sequence of fully reversed, sinusoidal cycles at a frequency equal to $1/(1.5T_1)$.
 - a. Ten fully reversed cycles at the displacement in the energy-dissipation device corresponding to 0.33 times the MCE_R device displacement;
 - b. Five fully reversed cycles at the displacement in the energy-dissipation device corresponding to 0.67 times the MCE_R device displacement;
 - c. Three fully reversed cycles at the displacement in the energy-dissipation device corresponding to 1.0 times the MCE_R device displacement; and
 - d. Where test (c) produces a force in the energy-dissipation device that is less than the MCE_R force in the device from analysis, test (c) shall be repeated at a frequency that produces a force equal to or greater than the MCE_R force from analysis.
3. Where the damping device characteristics vary with operating temperature, the tests of Section 18.6.1.2, 2(a) to 2(d) shall be conducted on at least one device, at a minimum of two additional temperatures (minimum and maximum), that bracket the design temperature range.

EXCEPTION: Damping devices are permitted to be tested by alternative methods provided that all of the following conditions are met:

- a. Alternative methods of testing are equivalent to the cyclic testing requirements of this section.
- b. Alternative methods capture the dependence of the damping device response on ambient temperature,

frequency of loading, and temperature rise during testing.

- c. Alternative methods are approved by the RDP responsible for the design of the structure.
4. If the force-deformation properties of the damping device at any displacement less than or equal to the maximum device displacement change by more than 15% for changes in testing frequency from $1/(1.5T_1)$ to $2.5/T_1$, then the preceding tests (2(a) through 2(c)) shall also be performed at frequencies equal to $1/T_1$ and $2.5/T_1$.

EXCEPTION: When full-scale dynamic testing is not possible because of test machine limitations, it is permitted to use reduced-scale prototypes to qualify the rate-dependent properties of damping devices, provided that scaling principles and similitude are used in the design of the reduced-scale devices and the test protocol.

18.6.1.3 Testing Similar Devices. Prototype tests need not be performed on a particular damping device if there exists a previously prototype-tested unit that meets all of the following conditions:

1. It is of similar dimensional characteristics, internal construction, and static and dynamic internal pressures (if any) to the subject damping device; and
2. It is of the same type and materials as the subject damping device; and
3. It was fabricated using identical documented manufacturing and quality control procedures that govern the subject damping device; and
4. It was tested under similar maximum strokes and forces to those required of the subject damping device.

18.6.1.4 Determination of Force-Velocity-Displacement Characteristics. The force-velocity-displacement characteristics of the prototype damping device shall be based on the cyclic displacement tests specified in Section 18.6.1.2 and all of the following requirements:

1. The maximum force and minimum force at zero displacement, the maximum force and minimum force at maximum device displacement, and the area of hysteresis loop (E_{loop}) shall be calculated for each cycle of deformation. Where required, the effective stiffness of a damping device shall be calculated for each cycle of deformation using Eq. (17.8-1).
2. Damping device nominal test properties for analysis and design shall be based on the average value for the first three cycles of test at a given displacement. For each cycle of each test, corresponding lambda factors (λ_{test}) for cyclic effects shall be established by comparison of nominal and per-cycle properties.
3. Lambda (λ) factors for velocity and temperature shall be determined simultaneously with those for cyclic effects where full-scale prototype test data are available. Where these or similar effects are determined from separate tests, lambda factors shall be established by comparison of properties determined under prototype test conditions with corresponding properties determined under the range of test conditions applicable to the property variation parameter.

18.6.1.5 Device Adequacy. The performance of a prototype damping device shall be deemed adequate if all of the conditions listed below are satisfied. The 15% limits specified in the following text are permitted to be increased by the RDP

responsible for the design of the structure provided that the increased limit has been demonstrated by analysis not to have a deleterious effect on the response of the structure.

18.6.1.5.1 Displacement-Dependent Damping Devices. The performance of the prototype displacement-dependent damping devices shall be deemed adequate if all of the following conditions, based on tests specified in Section 18.6.1.2, are satisfied:

1. For Test 1, no signs of damage including leakage, yielding, or breakage.
2. For Tests 2, 3, and 4, the maximum force and minimum force at zero displacement for a damping device for any one cycle does not differ by more than 15% from the average maximum and minimum forces at zero displacement as calculated from all cycles in that test at a specific frequency and temperature.
3. For Tests 2, 3, and 4, the maximum force and minimum force at maximum device displacement for a damping device for any one cycle does not differ by more than 15% from the average maximum and minimum forces at the maximum device displacement as calculated from all cycles in that test at a specific frequency and temperature.
4. For Tests 2, 3, and 4, the area of hysteresis loop (E_{loop}) of a damping device for any one cycle does not differ by more than 15% from the average area of the hysteresis loop as calculated from all cycles in that test at a specific frequency and temperature.
5. The average maximum and minimum forces at zero displacement and maximum displacement, and the average area of the hysteresis loop (E_{loop}), calculated for each test in the sequence of Tests 2, 3, and 4, shall not differ by more than 15% from the target values specified by the RDP responsible for the design of the structure.
6. The average maximum and minimum forces at zero displacement and the maximum displacement, and the average area of the hysteresis loop (E_{loop}), calculated for Test 2(c) shall fall within the limits specified by the RDP, as described by the nominal properties and the lambda factor for specification tolerance ($\lambda_{(spec, max)}$ and $\lambda_{(spec, min)}$) from Section 18.2.4.5.
7. The test lambda factors for damping units, determined in accordance with Section 18.6.1.4, shall not exceed the values specified by the RDP in accordance with Section 18.2.4.5.

18.6.1.5.2 Velocity-Dependent Damping Devices. The performance of the prototype velocity-dependent damping devices shall be deemed adequate if all of the following conditions, based on tests specified in Section 18.6.1.2, are satisfied:

1. For Test 1, no signs of damage including leakage, yielding, or breakage.
2. For velocity-dependent damping devices with stiffness, the effective stiffness of a damping device in any one cycle of Tests 2, 3, and 4 does not differ by more than 15% from the average effective stiffness as calculated from all cycles in that test at a specific frequency and temperature.
3. For Tests 2, 3, and 4, the maximum force and minimum force at zero displacement for a damping device for any one cycle does not differ by more than 15% from the average maximum and minimum forces at zero displacement as calculated from all cycles in that test at a specific frequency and temperature.
4. For Tests 2, 3, and 4, the area of hysteresis loop (E_{loop}) of a damping device for any one cycle does not differ by more

than 15% from the average area of the hysteresis loop as calculated from all cycles in that test at a specific frequency and temperature.

5. The average maximum and minimum forces at zero displacement, effective stiffness (for damping devices with stiffness only), and average area of the hysteresis loop (E_{loop}), calculated for Test 2(c), shall fall within the limits specified by the RDP, as described by the nominal properties and the lambda factor for specification tolerance ($\lambda_{(spec, max)}$ and $\lambda_{(spec, min)}$) from Section 18.2.4.5.
6. The test lambda factors for damping units determined in accordance with Section 18.6.1.4 shall not exceed the values specified by the RDP in accordance with Section 18.2.4.5.

18.6.2 Production Tests. Prior to installation in a building, damping devices shall be tested in accordance with the requirements of this section.

A test program for the production damping devices shall be established by the RDP. The test program shall validate the nominal properties by testing 100% of the devices for three cycles at 0.67 times the MCE_R stroke at a frequency equal to $1/(1.5T_1)$. The measured values of the nominal properties shall fall within the limits provided in the project specifications. These limits shall agree with the specification tolerances on nominal design properties established in Section 18.2.4.5.

EXCEPTION: Production damping devices need not be subjected to this test program if it can be shown by other means that their properties meet the requirements of the project specifications. In such cases, the RDP shall establish an alternative program to ensure the quality of the installed damping devices. This alternative program shall include production testing of at least one device of each type and size, unless project-specific prototype tests have been conducted on that identical device type and size. Devices that undergo inelastic action or are otherwise damaged during this test shall not be used in construction.

18.7 ALTERNATE PROCEDURES AND CORRESPONDING ACCEPTANCE CRITERIA

Structures analyzed by the response spectrum procedure shall meet the requirements of Sections 18.7.1, 18.7.3, and 18.7.4. Structures analyzed by the equivalent lateral force procedure shall meet the requirements of Sections 18.7.2, 18.7.3, and 18.7.4.

18.7.1 Response Spectrum Procedure. Where the response spectrum procedure is used to analyze a structure with a damping system, the requirements of this section shall apply.

18.7.1.1 Modeling. A mathematical model of the seismic force-resisting system and damping system shall be constructed that represents the spatial distribution of mass, stiffness, and damping throughout the structure. The model and analysis shall comply with the requirements of Section 12.9 for the seismic force-resisting system and to the requirements of this section for the damping system. The stiffness and damping properties of the damping devices used in the models shall be based on or verified by testing of the damping devices as specified in Section 18.6.

The elastic stiffness of elements of the damping system other than damping devices shall be explicitly modeled. Stiffness of damping devices shall be modeled depending on damping device type as follows:

1. For displacement-dependent damping devices: Displacement-dependent damping devices shall be modeled with an effective stiffness that represents damping device force at

the response displacement of interest (e.g., design story drift). Alternatively, the stiffness of hysteretic and friction damping devices is permitted to be excluded from response spectrum analysis provided that design forces in displacement-dependent damping devices, Q_{DSD} , are applied to the model as external loads (Section 18.7.4.5).

2. For velocity-dependent damping devices: Velocity-dependent damping devices that have a stiffness component (e.g., viscoelastic damping devices) shall be modeled with an effective stiffness corresponding to the amplitude and frequency of interest.

18.7.1.2 Seismic Force-Resisting System

18.7.1.2.1 Seismic Base Shear. The seismic base shear, V , of the structure in a given direction shall be determined as the combination of modal components, V_m , subject to the limits of Eq. (18.7-1):

$$V \geq V_{min} \quad (18.7-1)$$

The seismic base shear, V , of the structure shall be determined by the square root of the sum of the squares method (SRSS) or complete quadratic combination of modal base shear components, V_m .

18.7.1.2.2 Modal Base Shear. Modal base shear of the m th mode of vibration, V_m , of the structure in the direction of interest shall be determined in accordance with Eqs. (18.7-2a) and (18.7-2b):

$$V_m = C_{Sm} \bar{W}_m \quad (18.7-2a)$$

$$\bar{W}_m = \frac{\left(\sum_{i=1}^n w_i \phi_{im} \right)^2}{\sum_{i=1}^n w_i \phi_{im}^2} \quad (18.7-2b)$$

where

C_{Sm} = seismic response coefficient of the m th mode of vibration of the structure in the direction of interest as determined from Section 18.7.1.2.4 ($m=1$) or Section 18.7.1.2.6 ($m > 1$);

$\bar{W}_m$ = effective seismic weight of the m th mode of vibration of the structure; and

ϕ_{im} = displacement amplitude at the i th level of the structure in the m th mode of vibration in the direction of interest, normalized to unity at the roof level.

18.7.1.2.3 Modal Participation Factor. The modal participation factor of the m th mode of vibration, Γ_m , of the structure in the direction of interest shall be determined in accordance with Eq. (18.7-3):

$$\Gamma_m = \frac{\bar{W}_m}{\sum_{i=1}^n w_i \phi_{im}} \quad (18.7-3)$$

18.7.1.2.4 Fundamental Mode Seismic Response Coefficient. The fundamental mode ($m=1$) seismic response coefficient, C_{S1} , in the direction of interest shall be determined in accordance with Eqs. (18.7-4) and (18.7-5):

For $T_{1D} < T_S$,

$$C_{S1} = \left(\frac{R}{C_d} \right) \frac{S_{DS}}{\Omega_0 B_{1D}} \quad (18.7-4)$$

For $T_{1D} \geq T_S$,

$$C_{S1} = \left(\frac{R}{C_d} \right) \frac{S_{D1}}{T_{1D}(\Omega_0 B_{1D})} \quad (18.7-5)$$

18.7.1.2.5 Effective Fundamental Mode Period Determination. The effective fundamental mode ($m=1$) period at the design earthquake ground motion, T_{1D} , and at the MCE_R ground motion, T_{1M} , shall be based on either explicit consideration of the postyield force deflection characteristics of the structure or determined in accordance with Eqs. (18.7-6) and (18.7-7):

$$T_{1D} = T_1 \sqrt{\mu_D} \quad (18.7-6)$$

$$T_{1M} = T_1 \sqrt{\mu_M} \quad (18.7-7)$$

18.7.1.2.6 Higher Mode Seismic Response Coefficient. Higher mode ($m > 1$) seismic response coefficient, C_{Sm} , of the m th mode of vibration ($m > 1$) of the structure in the direction of interest shall be determined in accordance with Eqs. (18.7-8) and (18.7-9):

For $T_m < T_S$,

$$C_{Sm} = \left(\frac{R}{C_d} \right) \frac{S_{DS}}{\Omega_0 B_{mD}} \quad (18.7-8)$$

For $T_m \geq T_S$,

$$C_{Sm} = \left(\frac{R}{C_d} \right) \frac{S_{D1}}{T_m(\Omega_0 B_{mD})} \quad (18.7-9)$$

where

T_m = period, in seconds, of the m th mode of vibration of the structure in the direction under consideration, and
 B_{mD} = numerical coefficient as set forth in Table 18.7-1 for effective damping equal to β_{mD} and period of the structure equal to T_m .

18.7.1.2.7 Design Lateral Force. Design lateral force at level i caused by the m th mode of vibration, F_{im} , of the structure in the direction of interest shall be determined in accordance with Eq. (18.7-10):

$$F_{im} = w_i \phi_{im} \frac{\Gamma_m}{W_m} V_m \quad (18.7-10)$$

Design forces in elements of the seismic force-resisting system shall be determined by the SRSS or complete quadratic combination of modal design forces.

18.7.1.3 Damping System. Design forces in damping devices and other elements of the damping system shall be determined on the basis of the floor deflection, story drift, and story velocity response parameters described in the following sections.

Displacements and velocities used to determine maximum forces in damping devices at each story shall account for the angle of orientation of each device from the horizontal and consider the effects of increased response caused by torsion required for design of the seismic force-resisting system.

Floor deflections at level i , δ_{iD} and δ_{iM} , story drifts, Δ_D and Δ_M , and story velocities, ∇_D and ∇_M , shall be calculated for both the design earthquake ground motions and the

MCE_R ground motions, respectively, in accordance with this section.

18.7.1.3.1 Design Earthquake Floor Deflection. The deflection of structure caused by the design earthquake ground motions at level i in the m th mode of vibration, δ_{imD} , of the structure in the direction of interest shall be determined in accordance with Eq. (18.7-11):

$$\delta_{imD} = D_{mD} \phi_{im} \quad (18.7-11)$$

The total design deflection at each floor of the structure shall be calculated by the SRSS or complete quadratic combination of modal design earthquake deflections.

18.7.1.3.2 Design Earthquake Roof Displacement. Fundamental ($m=1$) and higher mode ($m > 1$) roof displacements caused by the design earthquake ground motions, D_{1D} and D_{mD} , of the structure in the direction of interest shall be determined in accordance with Eqs. (18.7-12) and (18.7-13):

For $m=1$,

$$D_{1D} = \left(\frac{g}{4\pi^2} \right) \Gamma_1 \frac{S_{DS} T_{1D}^2}{B_{1D}} \geq \left(\frac{g}{4\pi^2} \right) \Gamma_1 \frac{S_{DS} T_1^2}{B_{1E}}, \quad T_{1D} < T_S \quad (18.7-12a)$$

$$D_{1D} = \left(\frac{g}{4\pi^2} \right) \Gamma_1 \frac{S_{D1} T_{1D}}{B_{1D}} \geq \left(\frac{g}{4\pi^2} \right) \Gamma_1 \frac{S_{D1} T_1}{B_{1E}}, \quad T_{1D} \geq T_S \quad (18.7-12b)$$

For $m > 1$,

$$D_{mD} = \left(\frac{g}{4\pi^2} \right) \Gamma_m \frac{S_{D1} T_m}{B_{mD}} \leq \left(\frac{g}{4\pi^2} \right) \Gamma_m \frac{S_{DS} T_m^2}{B_{mD}} \quad (18.7-13)$$

18.7.1.3.3 Design Earthquake Story Drift. Design story drift in the fundamental mode, Δ_{1D} , and higher modes, Δ_{mD} ($m > 1$), of the structure in the direction of interest shall be calculated in accordance with Section 12.8.6 using modal roof displacements of Section 18.7.1.3.2.

Total design story drift, ∇_D , shall be determined by the SRSS or complete quadratic combination of modal design earthquake drifts.

18.7.1.3.4 Design Earthquake Story Velocity. Design story velocity in the fundamental mode, ∇_{1D} , and higher modes, ∇_{mD} ($m > 1$), of the structure in the direction of interest shall be calculated in accordance with Eqs. (18.7-14) and (18.7-15):

$$\text{For } m=1, \quad \nabla_{1D} = 2\pi \frac{\Delta_{1D}}{T_{1D}} \quad (18.7-14)$$

$$\text{For } m > 1, \quad \nabla_{mD} = 2\pi \frac{\Delta_{mD}}{T_m} \quad (18.7-15)$$

Total design story velocity, ∇_D , shall be determined by the SRSS or complete quadratic combination of modal design velocities.

18.7.1.3.5 MCE_R Response. Total modal maximum floor deflection at level i , MCE_R story drift values, and MCE_R story

Table 18.7-1 Damping Coefficient, B_{V+I} , B_{1D} , B_{1E} , B_R , B_{1M} , B_{mD} , B_{mM} (Where Period of the Structure $\geq T_0$)

Effective Damping, β (percentage of critical)	B_{V+I} , B_{1D} , B_{1E} , B_R , B_{1M} , B_{mD} , B_{mM} (where period of the structure $\geq T_0$)
≤ 2	0.8
5	1.0
10	1.2
20	1.5
30	1.8
40	2.1
50	2.4
60	2.7
70	3.0
80	3.3
90	3.6
≥ 100	4.0

velocity values shall be based on Sections 18.7.1.3.1, 18.7.1.3.3, and 18.7.1.3.4, respectively, except design roof displacement shall be replaced by MCE_R roof displacement. MCE_R roof displacement of the structure in the direction of interest shall be calculated in accordance with Eqs. (18.7-16) to (18.7-17):

For $m = 1$,

$$D_{1M} = \left(\frac{g}{4\pi^2} \right) \Gamma_1 \frac{S_{MS} T_{1M}^2}{B_{1M}} \geq \left(\frac{g}{4\pi^2} \right) \Gamma_1 \frac{S_{MS} T_1^2}{B_{1E}}, \quad T_{1M} < T_S \quad (18.7-16a)$$

$$D_{1M} = \left(\frac{g}{4\pi^2} \right) \Gamma_1 \frac{S_{M1} T_{1M}}{B_{1M}} \geq \left(\frac{g}{4\pi^2} \right) \Gamma_1 \frac{S_{M1} T_1}{B_{1E}}, \quad T_{1M} \geq T_S \quad (18.7-16b)$$

For $m > 1$,

$$D_{mM} = \left(\frac{g}{4\pi^2} \right) \Gamma_m \frac{S_{M1} T_m}{B_{mM}} \leq \left(\frac{g}{4\pi^2} \right) \Gamma_m \frac{S_{MS} T_m^2}{B_{mM}} \quad (18.7-17)$$

where B_{mM} is a numerical coefficient as set forth in Table 18.7-1 for effective damping equal to β_{mM} and period of the structure equal to T_m .

18.7.2 Equivalent Lateral Force Procedure. Where the equivalent lateral force procedure is used to design a structure with a damping system, the requirements of this section shall apply.

18.7.2.1 Modeling. Elements of the seismic force-resisting system shall be modeled in a manner consistent with the requirements of Section 12.8. For purposes of analysis, the structure shall be considered to be fixed at the base.

Elements of the damping system shall be modeled as required to determine design forces transferred from damping devices to both the ground and the seismic force-resisting system. The effective stiffness of velocity-dependent damping devices shall be modeled.

Damping devices need not be explicitly modeled provided that effective damping is calculated in accordance with the procedures of Section 18.7.4 and used to modify response as required in Sections 18.7.2.2 and 18.7.2.3.

The stiffness and damping properties of the damping devices used in the models shall be based on or verified by testing of the damping devices as specified in Section 18.6.

18.7.2.2 Seismic Force-Resisting System

18.7.2.2.1 Seismic Base Shear. The seismic base shear, V , of the seismic force-resisting system in a given direction shall be determined as the combination of the two modal components, V_1 and V_R , in accordance with Eq. (18.7-18):

$$V = \sqrt{V_1^2 + V_R^2} \geq V_{\min} \quad (18.7-18)$$

where

V_1 = design value of the seismic base shear of the fundamental mode in a given direction of response, as determined in Section 18.7.2.2.2;

V_R = design value of the seismic base shear of the residual mode in a given direction, as determined in Section 18.7.2.2.6; and

$V_{\min}$ = minimum allowable value of base shear permitted for design of the seismic force-resisting system of the structure in the direction of interest, as determined in Section 18.2.1.1.

18.7.2.2.2 Fundamental Mode Base Shear. The fundamental mode base shear, V_1 , shall be determined in accordance with Eq. (18.7-19):

$$V_1 = C_{S1} \bar{W}_1 \quad (18.7-19)$$

where

C_{S1} = the fundamental mode seismic response coefficient, as determined in Section 18.7.2.2.4 and

$\bar{W}_1$ = the effective fundamental mode seismic weight, including portions of the live load as defined by Eq. (18.7-2b) for $m = 1$.

18.7.2.2.3 Fundamental Mode Properties. The fundamental mode shape, ϕ_{i1} , and participation factor, Γ_1 , shall be determined by either dynamic analysis using the elastic structural properties and deformational characteristics of the resisting elements or using Eqs. (18.7-20) and (18.7-21):

$$\phi_{i1} = \frac{h_i}{h_n} \quad (18.7-20)$$

$$\Gamma_1 = \frac{\bar{W}_1}{\sum_{i=1}^n w_i \phi_{i1}} \quad (18.7-21)$$

where

h_i = the height above the base to level i ;

h_n = the structural height as defined in Section 11.2; and

w_i = the portion of the total effective seismic weight, W , located at or assigned to level i .

The fundamental period, T_1 , shall be determined either by dynamic analysis using the elastic structural properties and deformational characteristics of the resisting elements, or using Eq. (18.7-22) as follows:

$$T_1 = 2\pi \sqrt{\frac{\sum_{i=1}^n w_i \delta_i^2}{g \sum_{i=1}^n f_i \delta_i}} \quad (18.7-22)$$

where

f_i = lateral force at level i of the structure distributed in accordance with Section 12.8.3 and

δ_i = elastic deflection at level i of the structure caused by applied lateral forces f_i .

18.7.2.2.4 Fundamental Mode Seismic Response Coefficient. The fundamental mode seismic response coefficient, C_{S1} , shall be determined using Eq. (18.7-23) or (18.7-24):

For $T_{1D} < T_S$,

$$C_{S1} = \left(\frac{R}{C_d} \right) \frac{S_{DS}}{\Omega_0 B_{1D}} \quad (18.7-23)$$

For $T_{1D} \geq T_S$,

$$C_{S1} = \left(\frac{R}{C_d} \right) \frac{S_{D1}}{T_{1D}(\Omega_0 B_{1D})} \quad (18.7-24)$$

where

S_{DS} = the design spectral response acceleration parameter in the short period range

S_{D1} = the design spectral response acceleration parameter at a period of 1 s; and

B_{1D} = numerical coefficient as set forth in Table 18.7-1 for effective damping equal to $\beta_{mD}(m=1)$ and period of the structure equal to T_{1D} .

18.7.2.2.5 Effective Fundamental Mode Period Determination. The effective fundamental mode period at the design earthquake, T_{1D} , and at the MCE_R , T_{1M} , shall be based on explicit consideration of the postyield force deflection characteristics of the structure or shall be calculated using Eqs. (18.7-25) and (18.7-26):

$$T_{1D} = T_1 \sqrt{\mu_D} \quad (18.7-25)$$

$$T_{1M} = T_1 \sqrt{\mu_M} \quad (18.7-26)$$

18.7.2.2.6 Residual Mode Base Shear. Residual mode base shear, V_R , shall be determined in accordance with Eq. (18.7-27):

$$V_R = C_{SR} \bar{W}_R \quad (18.7-27)$$

where

C_{SR} = the residual mode seismic response coefficient as determined in Section 18.7.2.2.8 and

$\bar{W}_R$ = the effective residual mode weight of the structure determined using Eq. (18.7-30).

18.7.2.2.7 Residual Mode Properties. Residual mode shape, ϕ_{iR} , participation factor, Γ_R , effective residual mode seismic weight of the structure, $\bar{W}_R$, and effective period, T_R , shall be determined using Eqs. (18.7-28) through (18.7-31):

$$\phi_{iR} = \frac{1 - \Gamma_1 \phi_{i1}}{1 - \Gamma_1} \quad (18.7-28)$$

$$\Gamma_R = 1 - \Gamma_1 \quad (18.7-29)$$

$$\bar{W}_R = W - \bar{W}_1 \quad (18.7-30)$$

$$T_R = 0.4T_1 \quad (18.7-31)$$

18.7.2.2.8 Residual Mode Seismic Response Coefficient. The residual mode seismic response coefficient, C_{SR} , shall be determined in accordance with Eq. (18.7-32):

$$C_{SR} = \left(\frac{R}{C_d} \right) \frac{S_{DS}}{\Omega_0 B_R} \quad (18.7-32)$$

where B_R is a numerical coefficient as set forth in Table 18.7-1 for effective damping equal to β_R , and period of the structure equal to T_R .

18.7.2.2.9 Design Lateral Force. The design lateral force in elements of the seismic force-resisting system at level i caused by fundamental mode response, F_{i1} , and residual mode response, F_{iR} , of the structure in the direction of interest shall be determined in accordance with Eqs. (18.7-33) and (18.7-34):

$$F_{i1} = w_i \phi_{i1} \frac{\Gamma_1}{W_1} V_1 \quad (18.7-33)$$

$$F_{iR} = w_i \phi_{iR} \frac{\Gamma_R}{W_R} V_R \quad (18.7-34)$$

Design forces in elements of the seismic force-resisting system shall be determined by taking the SRSS of the forces caused by fundamental and residual modes.

18.7.2.3 Damping System. Design forces in damping devices and other elements of the damping system shall be determined on the basis of the floor deflection, story drift, and story velocity response parameters described in the following sections.

Displacements and velocities used to determine maximum forces in damping devices at each story shall account for the angle of orientation of each device from the horizontal and consider the effects of increased response caused by torsion required for design of the seismic force-resisting system.

Floor deflections at level i , δ_{iD} and δ_{iM} , story drifts, Δ_D and Δ_M , and story velocities, ∇_D and ∇_M shall be calculated for both the design earthquake ground motions and the MCE_R ground motions, respectively, in accordance with the following sections.

18.7.2.3.1 Design Earthquake Floor Deflection. The total design deflection at each floor of the structure in the direction of interest shall be calculated as the SRSS of the fundamental and residual mode floor deflections. The fundamental and residual mode deflections caused by the design earthquake ground motions, δ_{i1D} and δ_{iRD} , at the center of rigidity of level i of the structure in the direction of interest shall be determined using Eqs. (18.7-35) and (18.7-36):

$$\delta_{i1D} = D_{1D} \phi_{i1} \quad (18.7-35)$$

$$\delta_{iRD} = D_{RD} \phi_{iR} \quad (18.7-36)$$

where

D_{1D} = fundamental mode design displacement at the center of rigidity of the roof level of the structure in the direction under consideration, Section 18.7.2.3.2, and

D_{RD} = residual mode design displacement at the center of rigidity of the roof level of the structure in the direction under consideration, Section 18.7.2.3.2.

18.7.2.3.2 Design Earthquake Roof Displacement. Fundamental and residual mode displacements caused by the design earthquake ground motions, D_{1D} and D_{1R} , at the center of rigidity of the roof level of the structure in the direction of interest shall be determined using Eqs. (18.7-37) and (18.7-38):

$$D_{1D} = \left(\frac{g}{4\pi^2} \right) \Gamma_1 \frac{S_{DS} T_{1D}^2}{B_{1D}} \geq \left(\frac{g}{4\pi^2} \right) \Gamma_1 \frac{S_{DS} T_1^2}{B_{1E}}, \quad T_{1D} < T_S \quad (18.7-37a)$$

$$D_{1D} = \left(\frac{g}{4\pi^2} \right) \Gamma_1 \frac{S_{D1} T_{1D}}{B_{1D}} \geq \left(\frac{g}{4\pi^2} \right) \Gamma_1 \frac{S_{D1} T_1}{B_{1E}}, \quad T_{1D} \geq T_S \quad (18.7-37b)$$

$$D_{RD} = \left(\frac{g}{4\pi^2} \right) \Gamma_R \frac{S_{D1} T_R}{B_R} \leq \left(\frac{g}{4\pi^2} \right) \Gamma_R \frac{S_{DS} T_R^2}{B_R} \quad (18.7-38)$$

18.7.2.3.3 Design Earthquake Story Drift. Design story drifts, Δ_D , in the direction of interest shall be calculated using Eq. (18.7-39):

$$\Delta_D = \sqrt{\Delta_{1D}^2 + \Delta_{RD}^2} \quad (18.7-39)$$

where

Δ_{1D} = design story drift caused by the fundamental mode of vibration of the structure in the direction of interest and
 Δ_{RD} = design story drift caused by the residual mode of vibration of the structure in the direction of interest.

Modal design story drifts, Δ_{1D} and Δ_{RD} , shall be determined as the difference of the deflections at the top and bottom of the story under consideration using the floor deflections of Section 18.7.2.3.1.

18.7.2.3.4 Design Earthquake Story Velocity. Design story velocities, ∇_D , in the direction of interest shall be calculated in accordance with Eqs. (18.7-40) through (18.7-42):

$$\nabla_D = \sqrt{\nabla_{1D}^2 + \nabla_{RD}^2} \quad (18.7-40)$$

$$\nabla_{1D} = 2\pi \frac{\Delta_{1D}}{T_{1D}} \quad (18.7-41)$$

$$\nabla_{RD} = 2\pi \frac{\Delta_{RD}}{T_R} \quad (18.7-42)$$

where

∇_{1D} = design story velocity caused by the fundamental mode of vibration of the structure in the direction of interest and
 ∇_{RD} = design story velocity caused by the residual mode of vibration of the structure in the direction of interest.

18.7.2.3.5 MCE_R Response. Total modal MCE_R floor deflections at level i , maximum story drifts, and maximum story velocities shall be based on the equations in Sections 18.7.2.3.1, 18.7.2.3.3, and 18.7.2.3.4, respectively, except that design roof displacements shall be replaced by MCE_R roof

displacements. MCE_R roof displacements shall be calculated in accordance with Eqs. (18.7-43) and (18.7-44):

$$D_{1M} = \left(\frac{g}{4\pi^2} \right) \Gamma_1 \frac{S_{MS} T_{1M}^2}{B_{1M}} \geq \left(\frac{g}{4\pi^2} \right) \Gamma_1 \frac{S_{MS} T_1^2}{B_{1E}}, \quad T_{1M} < T_S \quad (18.7-43a)$$

$$D_{1M} = \left(\frac{g}{4\pi^2} \right) \Gamma_1 \frac{S_{M1} T_{1M}}{B_{1M}} \geq \left(\frac{g}{4\pi^2} \right) \Gamma_1 \frac{S_{M1} T_1}{B_{1E}}, \quad T_{1M} \geq T_S \quad (18.7-43b)$$

$$D_{RM} = \left(\frac{g}{4\pi^2} \right) \Gamma_R \frac{S_{M1} T_R}{B_R} \leq \left(\frac{g}{4\pi^2} \right) \Gamma_R \frac{S_{MS} T_R^2}{B_R} \quad (18.7-44)$$

where

S_{M1} = the MCE_R, 5% damped, spectral response acceleration parameter at a period of 1 s adjusted for site class effects as defined in Section 11.4.4;

S_{MS} = the MCE_R, 5% damped, spectral response acceleration parameter at short periods adjusted for site class effects as defined in Section 11.4.4; and

B_{1M} = numerical coefficient as set forth in Table 18.7-1 for effective damping equal to β_{mM} ($m=1$) and period of structure equal to T_{1M} .

18.7.3 Damped Response Modification. As required in Sections 18.7.1 and 18.7.2, response of the structure shall be modified for the effects of the damping system.

18.7.3.1 Damping Coefficient. Where the period of the structure is greater than or equal to T_0 , the damping coefficient shall be as prescribed in Table 18.7-1. Where the period of the structure is less than T_0 , the damping coefficient shall be linearly interpolated between a value of 1.0 at a 0-second period for all values of effective damping and the value at period T_0 as indicated in Table 18.7-1.

18.7.3.2 Effective Damping. The effective damping at the design displacement, β_{mD} , and at the MCE_R displacement, β_{mM} , of the m th mode of vibration of the structure in the direction under consideration shall be calculated using Eqs. (18.7-45) and (18.7-46):

$$\beta_{mD} = \beta_I + \beta_{Vm} \sqrt{\mu_D} + \beta_{HD} \quad (18.7-45)$$

$$\beta_{mM} = \beta_I + \beta_{Vm} \sqrt{\mu_M} + \beta_{HM} \quad (18.7-46)$$

where

β_{HD} = component of effective damping of the structure in the direction of interest caused by postyield hysteretic behavior of the seismic force-resisting system and elements of the damping system at effective ductility demand, μ_D ;

β_{HM} = component of effective damping of the structure in the direction of interest caused by postyield hysteretic behavior of the seismic force-resisting system and elements of the damping system at effective ductility demand, μ_M ;

β_I = component of effective damping of the structure caused by the inherent dissipation of energy by elements of the structure, at or just below the effective yield displacement of the seismic force-resisting system;

β_{vm} = component of effective damping of the m th mode of vibration of the structure in the direction of interest caused by viscous dissipation of energy by the damping system, at or just below the effective yield displacement of the seismic force-resisting system;

μ_D = effective ductility demand on the seismic force-resisting system in the direction of interest caused by the design earthquake ground motions; and

μ_M = effective ductility demand on the seismic force-resisting system in the direction of interest caused by the MCE_R ground motions.

Unless analysis or test data support other values, the effective ductility demand of higher modes of vibration in the direction of interest shall be taken as 1.0.

18.7.3.2.1 Inherent Damping. Inherent damping, β_I , shall be based on the material type, configuration, and behavior of the structure and nonstructural components responding dynamically at or just below yield of the seismic force-resisting system. Unless analysis or test data support other values, inherent damping shall be taken as not greater than 3% of critical for all modes of vibration.

18.7.3.2.2 Hysteretic Damping. Hysteretic damping of the seismic force-resisting system and elements of the damping system shall be based either on test or analysis or shall be calculated using Eqs. (18.7-47) and (18.7-48):

$$\beta_{HD} = q_H(0.64 - \beta_I) \left(1 - \frac{1}{\mu_D} \right) \quad (18.7-47)$$

$$\beta_{HM} = q_H(0.64 - \beta_I) \left(1 - \frac{1}{\mu_M} \right) \quad (18.7-48)$$

where

q_H = hysteresis loop adjustment factor, as defined in Section 18.7.3.2.2.1;

μ_D = effective ductility demand on the seismic force-resisting system in the direction of interest caused by the design earthquake ground motions; and

μ_M = effective ductility demand on the seismic force-resisting system in the direction of interest caused by the MCE_R ground motions.

Unless analysis or test data support other values, the hysteretic damping of higher modes of vibration in the direction of interest shall be taken as zero.

18.7.3.2.2.1 Hysteresis Loop Adjustment Factor. The calculation of hysteretic damping of the seismic force-resisting system and elements of the damping system shall consider pinching and other effects that reduce the area of the hysteresis loop during repeated cycles of earthquake demand. Unless analysis or test data support other values, the fraction of full hysteretic loop area of the seismic force-resisting system used for design shall be taken as equal to the factor, q_H , calculated using Eq. (18.7-49):

$$q_H = 0.67 \frac{T_S}{T_1} \quad (18.7-49)$$

where

T_S = period defined by the ratio, S_{D1}/S_{DS} and

T_1 = period of the fundamental mode of vibration of the structure in the direction of interest.

The value of q_H shall not be taken as greater than 1.0 and need not be taken as less than 0.5.

18.7.3.2.3 Viscous Damping. Viscous damping of the m th mode of vibration of the structure, β_{vm} , shall be calculated using Eqs. (18.7-50) and (18.7-51):

$$\beta_{vm} = \frac{\sum_j W_{mj}}{4\pi W_m} \quad (18.7-50)$$

$$W_m = \frac{1}{2} \sum_j F_{im} \delta_{im} \quad (18.7-51)$$

where

W_{mj} = work done by j th damping device in one complete cycle of dynamic response corresponding to the m th mode of vibration of the structure in the direction of interest at modal displacements, δ_{im} ;

W_m = maximum strain energy in the m th mode of vibration of the structure in the direction of interest at modal displacements, δ_{im} ;

F_{im} = m th mode inertial force at level i ; and

δ_{im} = deflection of level i in the m th mode of vibration at the center of rigidity of the structure in the direction under consideration.

Viscous modal damping of displacement-dependent damping devices shall be based on a response amplitude equal to the effective yield displacement of the structure.

The calculation of the work done by individual damping devices shall consider orientation and participation of each device with respect to the mode of vibration of interest. The work done by individual damping devices shall be reduced as required to account for the flexibility of elements, including pins, bolts, gusset plates, brace extensions, and other components that connect damping devices to other elements of the structure.

18.7.3.3 Effective Ductility Demand. The effective ductility demand on the seismic force-resisting system caused by the design earthquake ground motions, μ_D , and caused by the MCE_R ground motions, μ_M , shall be calculated using Eqs. (18.7-52) to (18.7-54):

$$\mu_D = \frac{D_{1D}}{D_Y} \geq 1.0 \quad (18.7-52)$$

$$\mu_M = \frac{D_{1M}}{D_Y} \geq 1.0 \quad (18.7-53)$$

$$D_Y = \left(\frac{g}{4\pi^2} \right) \left(\frac{\Omega_0 C_d}{R} \right) \Gamma_1 C_{S1} T_1^2 \quad (18.7-54)$$

where

D_{1D} = fundamental mode design displacement at the center of rigidity of the roof level of the structure in the direction under consideration, Section 18.7.1.3.2 or 18.7.2.3.2;

D_{1M} = fundamental mode maximum displacement at the center of rigidity of the roof level of the structure in the direction under consideration, Section 18.7.1.3.5 or 18.7.2.3.5;

D_Y = displacement at the center of rigidity of the roof level of the structure at the effective yield point of the seismic force-resisting system;

R = response modification coefficient from Table 12.2-1;
 C_d = deflection amplification factor from Table 12.2-1;
 Ω_0 = overstrength factor from Table 12.2-1;
 Γ_1 = participation factor of the fundamental mode of vibration of the structure in the direction of interest, Section 18.7.1.2.3 or 18.7.2.2.3 ($m=1$);
 C_{S1} = seismic response coefficient of the fundamental mode of vibration of the structure in the direction of interest, Section 18.7.1.2.4 or 18.7.2.2.4 ($m=1$); and
 T_1 = period of the fundamental mode of vibration of the structure in the direction of interest.

The design ductility demand, μ_D , shall not exceed the maximum value of effective ductility demand, $\mu_{\max}$, given in Section 18.7.3.4.

EXCEPTION: It is permitted to use nonlinear modeling as described in Section 18.3 to develop a force-displacement (push-over) curve of the seismic force-resisting system. It is permitted to use this curve in lieu of the effective yield displacement, D_Y , of Eq. (18.7-54) to calculate the effective ductility demand caused by the design earthquake ground motions, μ_D , and caused by the MCE_R ground motions, μ_M , in Eqs. (18.7-52) and (18.7-53), respectively. In this case, the value of (R/C_d) shall be taken as 1.0 in Eqs. (18.7-4), (18.7-5), (18.7-8), and (18.7-9).

18.7.3.4 Maximum Effective Ductility Demand. For determination of the hysteresis loop adjustment factor, hysteretic damping, and other parameters, the maximum value of effective ductility demand, $\mu_{\max}$, shall be calculated using Eqs. (18.7-55) and (18.7-56):

For $T_{1D} \leq T_S$,

$$\mu_{\max} = 0.5[(R/(\Omega_0 I_e))^2 + 1] \quad (18.7-55)$$

For $T_1 \geq T_S$,

$$\mu_{\max} = R/(\Omega_0 I_e) \quad (18.7-56)$$

where

I_e = the Importance Factor determined in accordance with Section 11.5.1 and

T_{1D} = effective period of the fundamental mode of vibration of the structure at the design displacement in the direction under consideration.

For $T_1 < T_S < T_{1D}$, $\mu_{\max}$ shall be determined by linear interpolation between the values of Eqs. (18.7-55) and (18.7-56).

18.7.4 Seismic Load Conditions and Acceptance Criteria for RSA and ELF Procedures. Design forces and displacements determined in accordance with the response spectrum procedure of Section 18.7.1 or the equivalent lateral force procedure of Section 18.7.2 shall be checked using the strength design criteria of this standard and the seismic loading conditions of Section 18.7.4.3.

The seismic force-resisting system, damping system, seismic loading conditions, and acceptance criteria shall conform to the following subsections.

18.7.4.1 Seismic Force-Resisting System. The seismic force-resisting system shall satisfy the requirements of Section 12.2.1 using seismic base shear and design forces determined in accordance with Section 18.7.1.2 or 18.7.2.2.

The design story drift, Δ_D , as determined in either Section 18.7.1.3.3 or 18.7.2.3.3 shall not exceed (R/C_d) times the

allowable story drift, as obtained from Table 12.12-1, considering the effects of torsion as required in Section 12.12.1.

18.7.4.2 Damping System. The damping system shall satisfy the requirements of Section 12.2.1 for seismic design forces and seismic loading conditions determined in accordance with Section 18.7.4.3. Force-controlled elements of the damping system shall be designed for seismic forces that are increased by 20% from those corresponding to average MCE_R response.

18.7.4.3 Combination of Load Effects. The effects on the damping system and its components caused by gravity loads and seismic forces shall be combined in accordance with Section 12.4 using the effect of horizontal seismic forces, Q_E , determined in accordance with Section 18.7.4.5. The redundancy factor, ρ , shall be taken equal to 1.0 in all cases, and the seismic load effect including overstrength of Section 12.4.3 need not apply to the design of the damping system.

18.7.4.4 Modal Damping System Design Forces. Modal damping system design forces shall be calculated on the basis of the type of damping devices and the modal design story displacements and velocities determined in accordance with either Section 18.7.1.3 or 18.7.2.3.

Modal design story displacements and velocities shall be increased as required to envelop the total design story displacements and velocities determined in accordance with Section 18.3 where peak response is required to be confirmed by response history analysis.

For displacement-dependent damping devices: Design seismic force in displacement-dependent damping devices shall be based on the maximum force in the device at displacements up to and including the design story drift, Δ_D .

For velocity-dependent damping devices: Design seismic force in each mode of vibration in velocity-dependent damping devices shall be based on the maximum force in the device at velocities up to and including the design story velocity for the mode of interest.

Displacements and velocities used to determine design forces in damping devices at each story shall account for the angle of orientation of the damping device from the horizontal and consider the effects of increased floor response caused by torsional motions.

18.7.4.5 Seismic Load Conditions and Combination of Modal Responses. Seismic design force, Q_E , in each element of the damping system shall be taken as the maximum force of the following three loading conditions:

1. Stage of maximum displacement: Seismic design force at the stage of maximum displacement shall be calculated in accordance with Eq. (18.7-57):

$$Q_E = \Omega_0 \sqrt{\sum_m (Q_{mSFRS})^2} \pm Q_{DSD} \quad (18.7-57)$$

where

Q_{mSFRS} = force in an element of the damping system equal to the design seismic force of the m th mode of vibration of the structure in the direction of interest and

Q_{DSD} = force in an element of the damping system required to resist design seismic forces of displacement-dependent damping devices.

Seismic forces in elements of the damping system, Q_{DSD} , shall be calculated by imposing design forces of

displacement-dependent damping devices on the damping system as pseudostatic forces. Design seismic forces of displacement-dependent damping devices shall be applied in both positive and negative directions at peak displacement of the structure.

2. Stage of maximum velocity: Seismic design force at the stage of maximum velocity shall be calculated in accordance with Eq. (18.7-58):

$$Q_E = \sqrt{\sum_m (Q_{mDSV})^2} \quad (18.7-58)$$

where

Q_{mDSV} = force in an element of the damping system required to resist design seismic forces of velocity-dependent damping devices caused by the m th mode of vibration of the structure in the direction of interest.

Modal seismic design forces in elements of the damping system, Q_{mDSV} , shall be calculated by imposing modal design forces of velocity-dependent devices on the nondeformed damping system as pseudostatic forces. Modal seismic design forces shall be applied in directions consistent with the deformed shape of the mode of interest. Horizontal restraint forces shall be applied at each floor level i of the nondeformed damping system concurrent with the design forces in velocity-dependent damping devices such that the horizontal displacement at each level of the structure is zero. At each floor level i , restraint forces shall be proportional to and applied at the location of each mass point.

3. Stage of maximum acceleration: Seismic design force at the stage of maximum acceleration shall be calculated in accordance with Eq. (18.7-59):

$$Q_E = \sqrt{\sum_m (C_{mFD} \Omega_0 Q_{mSFRS} + C_{mFV} Q_{mDSV})^2} \pm Q_{DSD} \quad (18.7-59)$$

The force coefficients, C_{mFD} and C_{mFV} , shall be determined from Tables 18.7-2 and 18.7-3, respectively, using values of effective damping determined in accordance with the following requirements:

For fundamental-mode response ($m = 1$) in the direction of interest, the coefficients, C_{1FD} and C_{1FV} , shall be based on the velocity exponent, α , which relates device force to damping device velocity. The effective fundamental-mode damping shall be taken as equal to the total effective damping of the fundamental mode less the hysteretic component of damping ($\beta_{1D} - \beta_{HD}$ or $\beta_{1M} - \beta_{HM}$) at the response level of interest ($\mu = \mu_D$ or $\mu = \mu_M$).

For higher mode ($m > 1$) or residual-mode response in the direction of interest, the coefficients, C_{mFD} and C_{mFV} , shall be based on a value of α equal to 1.0. The effective modal damping shall be taken as equal to the total effective damping of the mode of interest (β_{mD} or β_{mM}). For determination of the coefficient C_{mFD} , the ductility demand shall be taken as equal to that of the fundamental mode ($\mu = \mu_D$ or $\mu = \mu_M$).

Table 18.7-2 Force Coefficient,^{a,b} C_{mFD}

Effective Damping	$\mu \leq 1.0$				$C_{mFD} = 1.0^c$
	$\alpha \leq 0.25$	$\alpha = 0.5$	$\alpha = 0.75$	$\alpha \geq 1.0$	
≤ 0.05	1.00	1.00	1.00	1.00	$\mu \geq 1.0$
0.1	1.00	1.00	1.00	1.00	$\mu \geq 1.0$
0.2	1.00	0.95	0.94	0.93	$\mu \geq 1.1$
0.3	1.00	0.92	0.88	0.86	$\mu \geq 1.2$
0.4	1.00	0.88	0.81	0.78	$\mu \geq 1.3$
0.5	1.00	0.84	0.73	0.71	$\mu \geq 1.4$
0.6	1.00	0.79	0.64	0.64	$\mu \geq 1.6$
0.7	1.00	0.75	0.55	0.58	$\mu \geq 1.7$
0.8	1.00	0.70	0.50	0.53	$\mu \geq 1.9$
0.9	1.00	0.66	0.50	0.50	$\mu \geq 2.1$
≥ 1.0	1.00	0.62	0.50	0.50	$\mu \geq 2.2$

^aUnless analysis or test data support other values, the force coefficient C_{mFD} for viscoelastic systems shall be taken as 1.0.

^bInterpolation shall be used for intermediate values of velocity exponent, α , and ductility demand, μ .

^c C_{mFD} shall be taken as equal to 1.0 for values of ductility demand, μ , greater than or equal to the values shown.

Table 18.7-3 Force Coefficient,^{a,b} C_{mFV}

Effective Damping	$\alpha \leq 0.25$	$\alpha = 0.5$	$\alpha = 0.75$	$\alpha \geq 1.0$
≤ 0.05	1.00	0.35	0.20	0.10
0.1	1.00	0.44	0.31	0.20
0.2	1.00	0.56	0.46	0.37
0.3	1.00	0.64	0.58	0.51
0.4	1.00	0.70	0.69	0.62
0.5	1.00	0.75	0.77	0.71
0.6	1.00	0.80	0.84	0.77
0.7	1.00	0.83	0.90	0.81
0.8	1.00	0.90	0.94	0.90
0.9	1.00	1.00	1.00	1.00
≥ 1.0	1.00	1.00	1.00	1.00

^aUnless analysis or test data support other values, the force coefficient C_{mFV} for viscoelastic systems shall be taken as 1.0.

^bInterpolation shall be used for intermediate values of velocity exponent, α .

18.7.4.6 Inelastic Response Limits. Elements of the damping system are permitted to exceed strength limits for design loads provided it is shown by analysis or test that each of the following conditions are satisfied:

1. Inelastic response does not adversely affect damping system function.
2. Element forces calculated in accordance with Section 18.7.4.5, using a value of Ω_0 taken as equal to 1.0, do not exceed the strength required to satisfy the load combinations of Section 12.4.

18.8 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

See Chapter 23 for the list of consensus standards and other documents that shall be considered part of this standard to the extent referenced in this chapter.